

Breathero: Not Another Slow Breathing Game — Exploring Faster-Paced VR Breathing Exercise Games

Shun-Yu, Wang*
National Taiwan University
Taipei City, Taiwan
r11944013@csie.ntu.edu.tw

Chan Weng IO
National Taiwan University of
Science and Technology
Taipei City, Taiwan
m11110002@gapps.ntust.edu.tw

Chia-Yu, Cheng*
National Taiwan University
Taipei City, Taiwan
r11922117@ntu.edu.tw

Yu-Hsuan, Huang
National Taiwan University of
Science and Technology
Taipei City, Taiwan
m11110112@mail.ntust.edu.tw

Shu-Meng, Huang*
National Taiwan University
Taipei City, Taiwan
r11944010@csie.ntu.edu.tw

Jhih-Wei, Lin
National Taiwan Normal University
Taipei City, Taiwan
weiwei20021007@gmail.com



Figure 1: *Breathero* is a faster-paced VR game that uses breathing techniques for main interaction, which contains (A) Fire Breathing, (B) Box Breathing, and (C) Full Breathing.

ABSTRACT

BreatHero is a faster-paced Virtual Reality (VR) action game that explores the potential of combining breathing techniques learning with the combat elements of action games. *BreatHero* uses breathing techniques as various skills for attacking and defending against the enemies, so players can do breathing exercises while enjoying defeating the in-game enemies. We choose three breathing techniques: *Kapalabhati*, *Box Breathing*, and *Full Yogic Breathing*, which are all beneficial for physical and mental health. To enhance breathing techniques learning and memorization in action games, we present two design concepts: *Imitative Breathing Feedback* and *Associate Learning on Breathing Techniques*. Results from our study indicated that players found the game to be more engaging and were willing

*These authors contributed equally to this research.

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to play it repeatedly, suggesting that breathing exercises can be effectively integrated into faster-paced action games.

CCS CONCEPTS

• **Human-centered computing** → **Virtual reality**; • **Applied computing** → **Computer games**; • **Software and its engineering** → **Interactive games**.

KEYWORDS

Breathing exercises, Virtual Reality, Breathing game, Breath detection

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1 INTRODUCTION

Recently, due to the impact of Covid-19, there has been an increasing awareness of the importance of lung health, and breathing

exercises have become popular as they are simple and easy to perform. There are many different types of breathing techniques, each with specific patterns, rhythms, and timing requirements. Long-term practice of these techniques can improve physical and mental well-being, such as enhancing lung function and regulating emotion. In the past, there have been several studies that have used breathing as an input in games [6, 9–11], and even the development of VR games for doing breathing exercises [3, 5, 7, 12]. These games, however, are often slow-paced and lack the replayability, which limits their ability to attract players for long-term playing to reach the benefits of breathing training. Therefore, there is research has shown that using breathing as the game input in a faster-paced VR First Person Shooter (FPS) game can effectively increase player immersion and engagement [10]. However, these studies have only used breathing units (e.g., breathing-out), as the main interaction in the game, and were not designed for breathing exercises. In this regard, **we proposed *BreatHero*, a faster-paced VR action game designed to provide players with a fun and engaging way to do breathing exercises**, as shown in Figure 1. By leveraging the inherent motivation in the VR action game, we aim to achieve long-term training that improves lung function or overall physical health.

Three recommended breathing techniques were selected for our game: 1) *Kapalabhati* [8], a fast-paced technique aimed at increasing heart rate and uplifting the mind, 2) *Box Breathing* [1], a slow-paced technique aimed at reducing stress and improving concentration, and 3) *Full Yogic Breathing* [2], a deep breathing technique aimed at improving lung function. These techniques were then designed for the game as three distinct skills: 1) *Fire Breath* as an attacking skill, 2) *Box Breath* as a freezing skill, and 3) *Full Breath* as a defensive skill.

To achieve the goal of breath training, two breathing training concepts for action games have been established in *BreatHero*. The concepts include:

- **Imitative Breathing Feedback:** *BreatHero* provides visual cues to indicate the current phase of breathing skills. This design concept is based on one of the design guidelines in "Life Tree" [5] which incorporates breath feedback that creates a sense of synchrony between the player's breath and in-game objects, visualization, or sound. Thereby enhancing the playful experience and the efficiency of breathing exercises in games.
- **Associate Learning on Breathing Techniques:** *BreatHero* employs a paired associate learning approach which is widely used in verbal memorization [4], and we extend it to breathing technique memorization. In our approach, each breathing technique is used to deal with a specific in-game enemy, so it makes the enemy a stimulus for players to recall the corresponding breathing technique.

To evaluate the player experience of *BreatHero*, we conducted an 8-person study to investigate the suitability of combining breathing exercises with faster-paced VR action games. The results of the study showed that users found the faster-paced VR action game to be more interesting and felt that they had learned breathing techniques. In conclusion, this study suggests that breath training

is not limited to slow-paced VR games, but can also be effectively combined into faster-paced VR action games.

2 RELATED WORK

2.1 Breathing as Input to Digital Games

Several studies have explored using breathing as the controlled interface in VR action games [3, 10], of which *BreathVR* [10] is the first one proposed and presents the game design strategies for breathing actions. Its interface is based on three breathing-out actions and one breathing-held action mapped into different in-game skills. For the non-VR breathing-based games [6, 9, 11], they detect breathing-in, breathing-out, or breathing rate to control the in-game objects. These games have proposed methods for using breathing as game input. However, their approach to breathing actions was based on an intuitive design of strength and duration. In contrast, for our breathing action game, we have chosen to incorporate popular yogic breathing techniques, which contained breathing-in, breathing-out, and breath-held, as the core skills in our game.

2.2 Breathing-based Meditation VR Games

Breathing-based VR games for health promotion are usually slow-paced meditation games that help players relieve anxiety [7, 12]. However, these games do not guide players on how to breathe using specific techniques. In contrast, *Life Tree* [5] guides players to practice pursed-lip breathing, and the in-game tree grows and becomes more colorful if the player follows the correct breathing rhythm. While slow-paced meditation games can help players to focus on their breathing, they lack the replayability. As breathing exercises are a long-term training process, we designed a faster-paced action game that allows players to learn and practice multiple breathing skills with great replayability.

3 GAME CONCEPT AND DESIGN

In *BreatHero*, we have included three different breathing techniques, known as 1) "*Kapalabhati*" [8], which involves inhaling one time and exhaling four times in rapid succession while contracting the abdomen during exhalation. 2) "*Box Breathing*" [1], which involves four steps: breathing in, holding the breath, breathing out, and holding the breath again, with each step lasting for four seconds. and 3) "*Full Yogic Breathing*" [2], which involves inhalation while raising the arms and exhalation while lowering the arms. These three techniques correspond to three different skills in the game.

3.1 Imitative Breathing Feedback

Imitative Breathing Feedback, one of the design guidelines from "Life Tree" [5], they mentioned that "*Breathing feedback designed in a way which allows players to feel as if the objects in the game are imitating their breathing pattern in a playful way.*" Furthermore, they also suggested a concept, *Non-interruptive breathing feedback during gameplay*, which is to integrate breathing feedback into gameplay without disrupting the player's immersion in the game, while simultaneously promoting the player's awareness of their

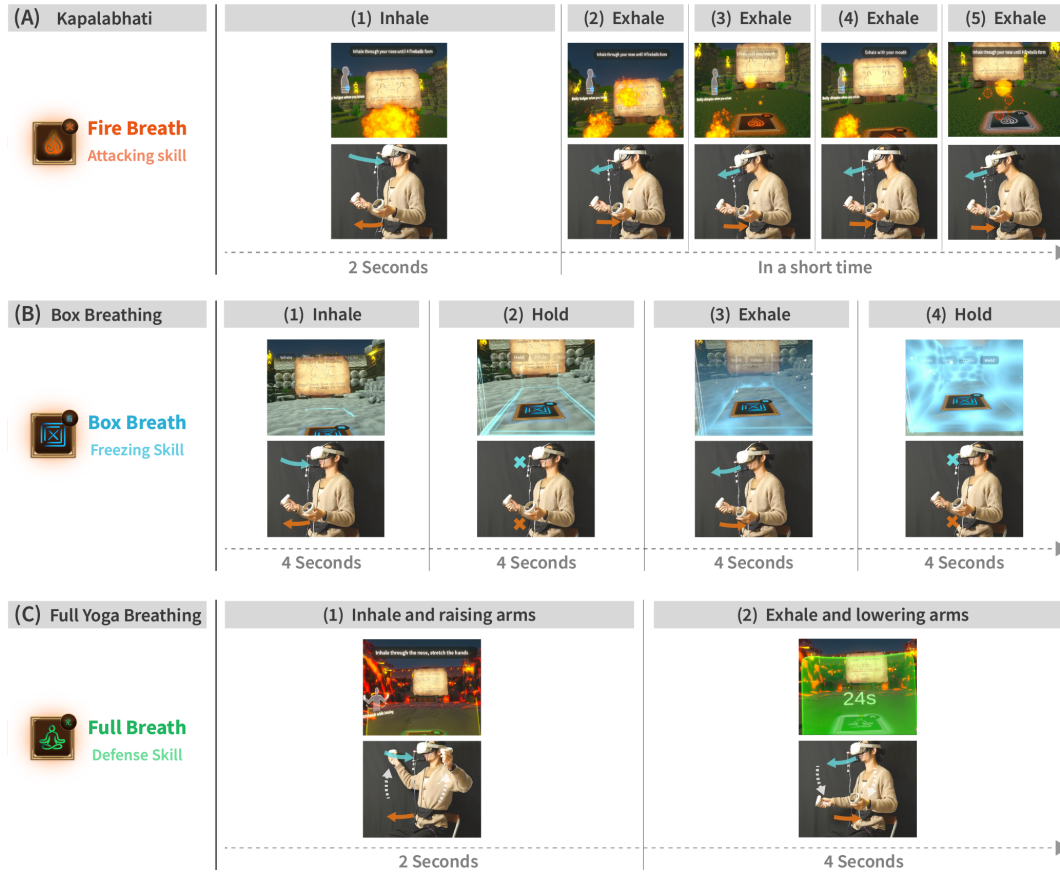


Figure 2: The Steps of (A) *Fire Breathing*, (B) *Box Breathing*, and (C) *Full Breathing*.

breathing patterns. Following these guidelines, we designed visualized feedback for the three breathing skills, which are introduced below, as shown in Figure 2.

- **"Fire Breath"**: An attacking skill, which allows players to shoot fireballs to defeat the sprites. As Figure 2(A), this skill is associated with the "*Kapalabhati*". To use this skill in the game, the player must inhale first, and the fireball will gradually grow in front of the player. Once the fireball splits into four smaller fireballs, the player must exhale four times to shoot the four fireballs.
- **"Box Breath"**: A freezing skill consists of four steps, which allows players to draw a box that can freeze the sprites inside. As in Figure 2(B), this skill is associated with "*Box Breathing*". In the first step, the player inhales and a square frame appears on the ground. Once the frame is fully formed, the player must hold their breath for the first time, and four pillars extend from the corners of the square. As the pillars finish extending, the color would change, providing a visual cue to start exhaling. As the player exhales, the space between the pillars is filled, forming a complete cube space. Finally, the player holds his breath for the second time, causing the cube space to shatter, freezing any sprites within it.

- **"Full Breath"**: A defensive skill that allows players to create a shield that can protect them from sprites' fire attacks. As Figure 2(C). As the player inhales and raises their arms, a yellow shield will float in front of them, growing until it is fully formed and shrink. Upon exhalation and lowering the arms, the shield will turn green and a 24-second countdown timer will appear on the shield while exhalation is done, indicating the shield's remaining time.

3.2 Associate Learning on Breathing Techniques

To help players memorize different breathing skills, *BreatHero* employed a paired associate learning approach, where each breathing technique was used to deal with a specific in-game sprite. Therefore, we designed three different elements of sprites, which were "tree", "stone" and "fire", as Figure 3. This approach also became our game mechanics: Players switched between breathing skills according to the sprite they were facing. This design made the sprite a stimulus for players to recall the corresponding breathing skill.

- **Tree sprite** has four visible weaknesses, represented by red circles, that players must target using *Fire Breath*.
- **Stone sprite** is able to dodge attacks by flashing to different locations. Therefore, players must use *Box Breath* to freeze the sprites before attacking them with *Fire Breath*.



Figure 3: Associate Learning on Breathing Techniques

- **Fire sprite** becomes invulnerable for a period of 20 seconds as it charges up a fire attack. Players must use *Full Breath* to create a shield that will protect them from the subsequent eruption of fire.
- **Boss sprite** possesses all the elements of previous sprites and alternates between them during the battle, requiring players to apply their skills accordingly.

3.3 Game Flow

The game comprises three tutorial levels, each lasting approximately three minutes and followed by boss level lasting five minutes. Therefore, the total playing time was approximately 15 minutes. In each of the first three levels, players are introduced to a distinct breathing technique and practice them by defeating various in-game sprites. In the final level, players face the boss which can switch among three elements: tree, stone, and fire. The glowing object (e.g., tree, stone, or red horn) on the boss's head indicates its current element. The boss also summons minions at critical moments to increase the challenge. Players can freely strategize on how to defeat the boss and its minions, then finally save the world, becoming a breathing hero.

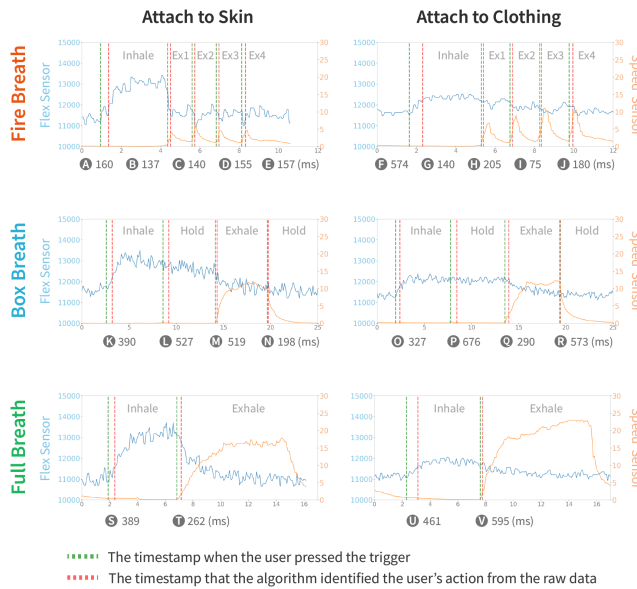


Figure 4: The evaluation of the breathing detection.

4 IMPLEMENTATION

4.1 Hardware Sensor

To accurately detect the player's breathing and instruct the player to breathe, we installed a sensor on the VR headset that can measure the wind speed, called the "Wind Sensor Rev. C.". It utilizes a thermal anemometer based on a traditional method called "hot-wire," where a component is heated to a constant temperature, and the electrical power required to maintain the constant temperature is measured as the wind changes, then wind speed can be calculated. However, we cannot determine whether the player is inhaling or exhaling only based on wind speed. To address this, we also attach a Flex Sensor to the player's abdomen. As the player performs diaphragmatic breathing, the sensor will change its resistance value as the abdomen is inflated or deflated, allowing us to detect whether the player is inhaling or exhaling. Additionally, the player will wear a waist belt over the flex sensor to increase the accuracy of the flex sensor's detection. By combining these two sensors, we can determine the player's breathing pattern, including inhaling, exhaling, or holding their breath, as well as the duration of the breath. Finally, we will connect these two sensors to the computer through an Arduino Uno board, as shown in Figure 5.

4.2 Evaluation

To evaluate the performance of combining two sensors, we did an analysis in which we attached the flex sensor to the abdomen of one of the authors, who performed each of the three breathing techniques ten times. To assess the difference between attaching the flex sensor directly to the skin or clothing, we also attached the flex sensor to the author's clothes and repeated the experiment. During each trial, the author pressed the trigger on the controller while performing each breathing action (inhale, exhale, or hold) to measure latency and accuracy. Figure 4 shows the results of the analysis. The vertical green lines in the figures represent the average time when the user pressed the trigger, while red lines indicate the average time that the algorithm correctly identified the user's action from the raw data. The results showed that the average latency for inhaling and exhaling during the fire breath was 160ms, and for exhaling four times, it was 137ms, 140ms, 155ms, and 157ms. For the box breath, the average latency for inhaling, holding, exhaling, and holding again was 390ms, 527ms, 519ms, and 198ms, respectively. For the full breath, the average latency for inhaling and exhaling was 389ms and 262ms, respectively. When correctly performing diaphragmatic breathing, placing the flex sensor on the skin can effectively detect the user's breathing actions with larger

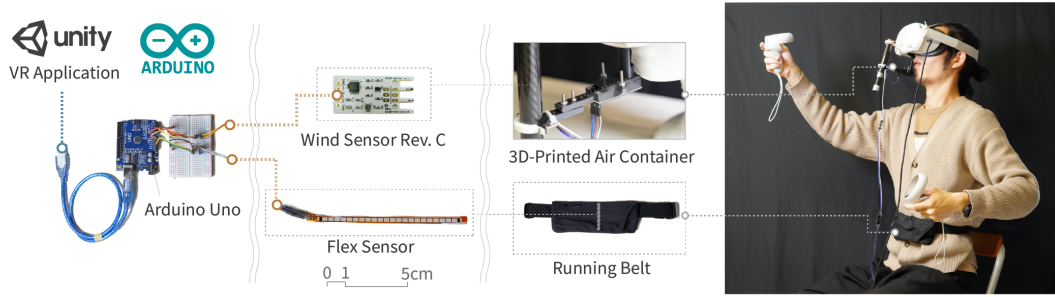


Figure 5: The integration of hardware and software in "BreatHero".

fluctuations in resistance, resulting in better accuracy than placing it on clothes.

5 USER STUDY

To investigate the feasibility of integrating respiratory training into faster-paced action games and the public's willingness to engage in long-term respiratory training through *BreatHero*. We conducted a user study.

We recruited 8 participants (4 males and 4 females, ages ranging from 20 to 27), half of whom had prior experience with respiratory training while the others had not. Additionally, one of the participants was a yoga teacher familiar with respiratory exercises. Participants were required to fill out a basic demographic form before the study began. They then sat on a chair and were helped to wear breath detection devices (Oculus VR headset with wind sensor and belt with abdominal bending sensor). A calibration procedure was performed to ensure the accuracy of the detection of breathing. After playing *BreatHero*, participants answered a three-stage questionnaire on a 7-point Likert Scale to assess "Game Experience", "Feasibility of *BreatHero*", and "Evaluation of Game Concept".

5.1 First Stage: Game Experience

In this stage, participants provided scores on four aspects of the game: "Replayability," "Immersion," "Entertainment," and "Fun," as shown in Figure 6. The 8 participants provided an average score of 5.125, 5.75, 5.875, and 5.75, respectively. Two participants even gave a score of 7 for the "Replayability," (P2, P7). They mentioned that the game was very entertaining and made them want to play again. This result suggests that users generally found the game entertaining and had the willingness to play it repeatedly in the context of the VR action game.

5.2 Second Stage: Feasibility of *BreatHero*

In this stage, participants were asked two questions. Q1: "Is breathing exercise suitable to be combined with VR action games?", Q2: "After playing *BreatHero*, how much confidence do you have in learning these three breathing techniques?"

As shown in Figure 7, the four participants without respiratory training experience gave an average score of 5.25 and 3.75 on these two questions, respectively. All of them believed that merging respiratory training with a VR action game could increase their willingness to engage in breathing exercises through gameplay. However,

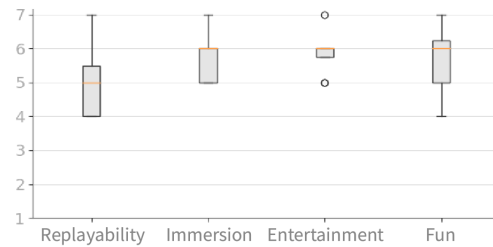


Figure 6: Game Experience.

as it was their first time playing *BreatHero*, they sometimes defeated sprites in a rush and did not focus on practicing breathing techniques, leading to the perception that they had not fully learned breathing techniques. They also noted that with more gameplay, they would become more proficient with the three techniques. The other four participants with respiratory training experience gave an average score of 2.75 and 4.25 on these two questions, respectively. Most of them reported that they had learned the three breathing techniques through the *BreatHero*. However, One(P2) of them, the yoga teacher, gave a score of 3, stating that "In general, breathing training is typically done by continuously practicing a specific technique until it is learned or effective, rather than switching breathing techniques based on the encountered elves." However, she still mentioned that "But switching breathing techniques makes this game more enjoyable", and suggested that "Maybe a specific breathing technique could be designed to defeat a specific sprite, rather than switching to the *Fire Breath* technique to attack them."

5.3 Third Stage: Evaluation of Game Concept

In this stage, participants were asked two questions to evaluate *Imitative Breathing Feedback*. Q1: "Did visualizations of skills help you learn the breathing technique?", Q2: "Did the visualizations of skills assist in your awareness of which stage of the breathing technique you were currently in?", and one question for *Associate Learning on Breathing Techniques*. Q3: "Did Associate Learning on Breathing Techniques help you memorize breathing techniques?"

As shown in Figure 8, the average score among eight participants for the *Imitative Breathing Feedback* was above 5 for the first two questions. Most participants supposed that this game design was effective in helping them learn breathing techniques and improve



Figure 7: Feasibility of BreatHero.

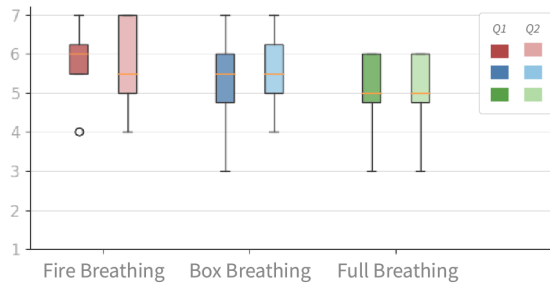


Figure 8: Imitative Breathing Feedback.

their immersion in the game. It also provided a clear indication of the current stage of breathing techniques. Additionally, as shown in Figure 8, the average score for the question on *Associate Learning on Breathing Techniques* was 4.625 with a median of 5. Most participants gave a score above 4, while a few participants did not notice the effect of this game design but still reported that they had successfully learned the three breathing techniques.

6 CRITICAL REFLECTION

BreatHero was designed to assist individuals in engaging in long-term respiratory training. Feedback from participants indicated that the game was enjoyable, immersive, and engaging. The majority of participants expressed positive feedback regarding the *Imitative Breathing Feedback* and *Associate Learning on Breathing Techniques* design, stating their willingness to play repeatedly and acknowledging its effectiveness in retaining breathing exercise techniques. For example, the visualization of breathing techniques can enhance the game's experience, aiding players in understanding when to inhale, exhale, or hold their breath.

However, there is a potential negative effect when players switch between different techniques. Some participants reported feeling overwhelmed when required to switch between different breathing techniques quickly while also trying to defeat sprites, leading to difficulties in completing the breathing exercises correctly. In order to reduce the negative effect, we suggest implementing specific breathing techniques in each tutorial levels to help users focus on learning one breathing technique one time. This approach would enhance the game's effectiveness in improving users' retention of breathing exercises and avoid the negative effects of switching between different techniques.

Moreover, balancing between "difficulty" and "maintaining the game's replayability" was one of the keys to successfully designing a faster-paced game. Hence, we also recommend implementing a gradual learning approach by allowing users to choose different difficulty levels based on their proficiency in breathing techniques, or gradually increasing the required breathing seconds to the correct breath duration in the early stages of the game. This approach should enable users to develop confidence in their ability to perform the breathing exercises and improve their performance in subsequent levels of the game and provide a more engaging and immersive experience for players.

In summary, allow players to learn breathing techniques smoothly and confidently through a progressive learning approach is important. Reducing the potential negative effect from the game mechanics as possible would make the game more accessible and effective for users engaging in long-term respiratory training.

7 CONCLUSION

In conclusion, our study investigated the feasibility of combining breathing exercises with faster-paced VR action games through the development of *BreatHero*. Our results positively suggest that the game concepts in *BreatHero* would increase players' engagement and retention of breathing techniques. Since not all players prefer slow-paced games, we believe that the incorporation of breathing exercises into faster-paced VR action games holds significant potential for long-term breath training.

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